

Question 1

The most recent iteration of the immersive theater experience *Sleep No More*, which premiered in New York City in 2011, transforms its performance space—a five-story warehouse—into a 1930s-era hotel. Audience members, who wander through the labyrinthine venue at their own pace and follow the actors as they play out simultaneous, interweaving narrative loops, confront the impossibility of experiencing the production in its entirety. The play's refusal of narrative coherence thus hinges on the sense of spatial fragmentation that the venue's immense and intricate layout generates.

What does the text most strongly suggest about *Sleep No More's* use of its performance space?

- A. The choice of a New York City venue likely enabled the play's creators to experiment with the use of theatrical space in a way that venues from earlier productions could not.
- B. Audience members likely find the experience of the play disappointing because they generally cannot make their way through the entire venue.
- C. The production's dependence on a particular performance environment would likely make it difficult to reproduce exactly in a different theatrical space.
- D. Audience members who navigate the space according to a recommended itinerary will likely have a better grasp of the play's narrative than audience members who depart from that itinerary.

Question 2

The following text is adapted from María Cristina Mena's 1914 short story "The Vine-Leaf."

It is a saying in the capital of Mexico that Dr. Malsufrido carries more family secrets under his hat than any archbishop.

The doctor's hat is, appropriately enough, uncommonly capacious, rising very high, and sinking so low that it seems to be supported by his ears and eyebrows, and it has a furry look, as if it had been brushed the wrong way, which is perhaps what happens to it if it is ever brushed at all. When the doctor takes it off, the family secrets do not fly out like a flock of parrots, but remain nicely bottled up beneath a dome of old and highly polished ivory.

Based on the text, how do people in the capital of Mexico most likely regard Dr. Malsufrido?

- A. Many have come to tolerate him despite his disheveled appearance.
- B. Few feel concerned that he will divulge their confidences.
- C. Some dislike how freely he discusses his own family.
- D. Most would be unimpressed by him were it not for his professional expertise.

Question 3

The following text is adapted from Countee Cullen's 1926 poem "Thoughts in a Zoo."

They in their cruel traps, and we in ours,
Survey each other's rage, and pass the hours
Commiserating each the other's woe,
To mitigate his own pain's fiery glow.
Man could but little proffer in exchange
Save that his cages have a larger range.
That lion with his lordly, untamed heart
Has in some man his human counterpart,
Some lofty soul in dreams and visions wrapped,
But in the stifling flesh securely trapped.

Based on the text, what challenge do humans sometimes experience?

- A. They cannot effectively tame certain wild animals because of a lack of compassion.
- B. They cannot focus on setting attainable goals because of a lack of motivation.
- C. They quickly become frustrated when faced with difficult tasks because of a lack of self-control.
- D. They have aspirations that cannot be fulfilled because of certain limitations.

Question 4

For centuries, the widespread acknowledgment of the involvement of the cerebellum—a dense brain structure in vertebrates—in coordinating motor control in humans has hindered recognition of other possible functions of the structure. Neuroscience research from the last two decades now suggests that the cerebellum regulates emotion and social behavior, and recent research by Ilaria Carta and colleagues has identified a pathway connecting the cerebellum to a center for motivation and reward processing known as the ventral tegmental area (VTA).

Which choice best states the main idea of the text?

- A. The recent verification of a pathway between the VTA and the cerebellum confirms the cerebellum's long-suspected role in motor coordination.
- B. Recent advances in the field of neuroscience have challenged widely accepted claims about the function of a pathway connecting the VTA and the cerebellum.
- C. The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers have been uncovering additional functions.
- D. Technological limitations have historically hindered the study of the cerebellum, but the recent development of new technologies has led to greater insights into its functions.

Question 5

Algae living within the tissues of corals play a critical role in keeping corals, and the marine ecosystems they are part of, thriving. Some coral species appear brown in color when healthy due to the algae colonies living in their tissues. In the event of an environmental stressor, the algae can die or be expelled, causing the corals to appear white. To recover the algae, the bleached corals then begin to produce bright colors, which block intense sunlight, encouraging the light-sensitive algae to recolonize the corals.

What does the text most strongly suggest about corals that produce bright colors?

- A. These corals have likely been subjected to stressful environmental conditions.
- B. These corals are likely more vulnerable to exposure from intense sunlight than white corals are.
- C. These corals have likely recovered from an environmental event without the assistance of algae colonies.
- D. These corals are more likely to survive without algae colonies than brown corals are.

Question 6

Having written the impassioned call to arms “Letter to the Spanish Americans” in 1791, Peruvian intellectual Juan Pablo Viscardo y Guzmán is often considered a forerunner for the independence movements in Latin America. But Viscardo’s role in history would have remained insignificant were it not for Venezuelan revolutionary Francisco de Miranda, who was handed the unpublished letter after Viscardo’s death. Miranda not only helped circulate the letter, but his edits and footnotes to the text position Miranda as a central figure in the text’s creation.

Which choice best states the main idea of the text?

- A. The original authorship of “Letter to the Spanish Americans” is disputed by contemporary historians.
- B. The majority of the most eloquently stated arguments in “Letter to the Spanish Americans” were written by Miranda.
- C. Miranda played a crucial role in influencing the content and distribution of “Letter to the Spanish Americans.”
- D. “Letter to the Spanish Americans” persuaded many people in Latin America to pursue national independence.

Question 7

In a paper about p-i-n planar perovskite solar cells (one of several perovskite cell architectures designed to collect and store solar power), Lyndsey McMillon-Brown et al. describe a method for fabricating the cell’s electronic transport layer (ETL) using a spray coating. Conventional ETL fabrication is accomplished using a solution of nanoparticles. The process can result in a loss of up to 80% of the solution, increasing the cost of manufacturing at scale—an issue that may be obviated by spray coating fabrication, which the researchers describe as “highly reproducible, concise, and practical.”

What does the text most strongly suggest about conventional ETL fabrication?

- A. It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.
- B. It is more expensive when manufacturing at scale than are processes for fabricating ETLs used in other perovskite solar cell architectures.
- C. It typically entails a greater loss of nanoparticle solution than do other established approaches for ETL fabrication.
- D. It is somewhat imprecise and therefore limits the potential effectiveness of p-i-n planar perovskite solar cells at capturing and storing solar power.

Question 8

Disco remains one of the most ridiculed popular music genres of the late twentieth century. But as scholars have argued, the genre is far less superficial than many people believe. Take the case of disco icon Donna Summer: she may have been associated with popular songs about love and heartbreak (subjects hardly unique to disco, by the way), but like many Black women singers before her, much of her music also reflects concerns about community and identity. These concerns are present in many of the genre’s greatest songs, and they generally don’t require much digging to reveal.

What does the text most strongly suggest about the disco genre?

- A. It has been unjustly ignored by most scholars despite the importance of the themes addressed by many of the genre’s songs.
- B. It evolved over time from a superficial genre focused on romance to a genre focused on more serious concerns.
- C. It has been unfairly dismissed for the inclusion of subject matter that is also found in other musical genres.
- D. It gave rise to a Black women’s musical tradition that has endured even though the genre itself faded in the late twentieth century.

Question 9

In a study of new technology adoption, Davit Marikyan et al. examined negative disconfirmation (which occurs when experiences fall short of one's expectations) to determine whether it could lead to positive outcomes for users. The team focused on established users of "smart home" technology, which presents inherent utilization challenges but tends to attract users with high expectations, often leading to feelings of dissonance. The researchers found that many users employed cognitive mechanisms to mitigate those feelings, ultimately reversing their initial sense of disappointment.

Which choice best states the main idea of the text?

- A. Research suggests that most users of smart home technology will not achieve a feeling of satisfaction given the utilization challenges of such technology.
- B. Although most smart home technology is aimed at meeting or exceeding users' high expectations, those expectations in general remain poorly understood.
- C. Research suggests that users with high expectations for a new technology can feel content with that technology even after experiencing negative disconfirmation.
- D. Although negative disconfirmation has often been studied, little is known about the cognitive mechanisms shaping users' reactions to it in the context of new technology adoption.

Question 10

Optical tweezers are specialized scientific tools—particularly useful in biology and medicine—that use high-powered beams of light to trap and manipulate minuscule particles for study. Use of the tool has led to several scientific and medical breakthroughs over the last few decades, but the particles are often under prolonged exposure to the intense heat of the light beams. To overcome the risk of overheating, and thereby damage, researchers sometimes attach nano-sized glass beads to particles, allowing the light to focus on the beads instead of the particles.

Based on the text, what is one advantage of attaching glass beads to particles when using optical tweezers?

- A. It decreases the time it takes for the optical tweezers to locate and capture the particles.
- B. It facilitates the maneuvering of particles without directly heating the particles themselves.
- C. It allows researchers to use weaker light beams to manipulate particles.
- D. It adds a material to which particles can transfer any heat absorbed from the optical tweezers' light beam.

Question 11

Some animal-behavior studies involve observing wild animals in their natural habitat, and some involve capturing wild animals and observing them in a laboratory. Each approach has advantages over the other. In wild studies, researchers can more easily presume that the animals are behaving normally, and in lab studies, researchers can more easily control factors that might affect the results. But if, for example, the results from a wild study and a lab study of Western scrub-jays (*Aphelocoma californica*) contradict each other, one or both of the studies must have failed to account for some factor that was relevant to the birds' behavior.

Which choice best states the main idea of the text?

- A. When the results of a natural-habitat study and those from a lab study of a wild animal such as the Western scrub-jay conflict, the study in the natural habitat is more likely than the lab study to have accurate results.
- B. Studying wild animals such as the Western scrub-jay in both their natural habitat and lab settings is likely to yield conflicting results that researchers cannot fully resolve.
- C. Wild animals such as the Western scrub-jay can be effectively studied in their natural habitat and in the lab, but each approach has drawbacks that could affect the accuracy of the findings.
- D. Differing results between natural-habitat and lab studies of wild animals such as the Western scrub-jay are a strong indication that both of the studies had design flaws that affected the accuracy of their results.

Question 12

Artificial leaves are a developing renewable energy technology that mimics the process of photosynthesis in plants. These devices are silicon-based solar cells coated in chemical catalysts that activate reactions that split water molecules into hydrogen and oxygen gas. The technology, while generating lots of interest, is not yet commercially viable as a large-scale energy source. To meet this challenge, scientists from many fields are researching ways to store, transport, and distribute the energy the devices produce while other scientists are working to improve the cost and efficiency of the devices.

Which choice best states the main idea of the text?

- Continued research and development in artificial-leaf technology is needed before the devices can be widely used as an
- A. energy source.
- The recent increase in the commercial use of artificial leaves as an energy source has encouraged many scientists to
- B. research ways to improve the technology.
- Artificial leaves split water molecules into oxygen and hydrogen gas using catalysts more efficiently than plants do using
- C. the process of photosynthesis.
- Artificial leaves were developed to mimic the natural process of photosynthesis in plants in order to store energy for long-
- D. term commercial use.

Question 13

Electronic music pioneer Wendy Carlos is credited with the music for three feature films: *A Clockwork Orange* (1971), *The Shining* (1980), and *Tron* (1982). However, her musical score for *A Clockwork Orange* is mostly made up of her arrangements of Ludwig van Beethoven's work. Also, almost all the music that she and Rachel Elkind composed for *The Shining* was unused by director Stanley Kubrick. It did not appear in the film. Of the three films, *Tron* is the one in which audiences can hear the most of Carlos's original compositions.

Which choice best states the main idea of the text?

- A. Of the three films for which Carlos is credited, *Tron* features the most original music from her.
- B. The director of *The Shining* used most of the music that Carlos composed for it.
- C. Beethoven is widely considered to be a more important composer than Carlos.
- D. Carlos is a notable innovator among film composers in the 1970s and 1980s.

Question 14

The following text is adapted from William Shakespeare's 1609 poem "Sonnet 27." The poem is addressed to a close friend as if he were physically present.

Weary with toil, I [hurry] to my bed,
The dear repose for limbs with travel tired;
But then begins a journey in my head
To work my mind, when body's work's expired:
For then my thoughts—from far where I abide—
[Begin] a zealous pilgrimage to thee,
And keep my drooping eyelids open wide,

What is the main idea of the text?

- A. The speaker is asleep and dreaming about traveling to see the friend.
- B. The speaker is planning an upcoming trip to the friend's house.
- C. The speaker is too fatigued to continue a discussion with the friend.
- D. The speaker is thinking about the friend instead of immediately falling asleep.

Question 15

The following text is from Maggie Pogue Johnson's 1910 poem "Poet of Our Race." In this poem, the speaker is addressing Paul Laurence Dunbar, a Black author.

Thou, with stroke of mighty pen,
Hast told of joy and mirth,
And read the hearts and souls of men
As cradled from their birth.
The language of the flowers,
Thou hast read them all,
And e'en the little brook
Responded to thy call.

Which choice best states the main purpose of the text?

- A. To praise a certain writer for being especially perceptive regarding people and nature
- B. To establish that a certain writer has read extensively about a variety of topics
- C. To call attention to a certain writer's careful and elaborately detailed writing process
- D. To recount fond memories of an afternoon spent in nature with a certain writer

Question 16

In eighteenth-century Britain, the aesthetic concept of the picturesque set a new standard for the appreciation of nature: in contrast with formal gardens, which were explicitly artificial, picturesque scenery—whose hallmarks were rugged terrains and medieval ruins—was prized for its appearance of untamed wilderness. Yet the experience of the picturesque was highly mediated: landscapes were admirable insofar as they adhered to depictions of nature in art and poetry, and tourists seeking the picturesque used optical devices, such as tinted mirrors, to transform views into painterly images.

Which choice best states the main idea of the text?

- (A) Although the picturesque gained currency as an aesthetic ideal in eighteenth-century Britain, the self-evident contradictions inherent in the concept drew much contemporary criticism.
- (B) Admirers of picturesque scenery in the eighteenth century were often unmindful of the extent to which their appreciation of nature was shaped by rapidly developing optical technologies.
- (C) The defining features of the picturesque landscape, including rugged terrains and medieval ruins, were highly valued by tourists primarily because of their relative prevalence in Britain.
- (D) Although the idea of the picturesque in nature was premised on the rejection of overt artifice, judgments about the picturesque were nevertheless deeply informed by principles derived from art.

Question 17

The following text is Vita Sackville-West's circa 1920 poem "Evening." Spars are ships' masts, moorings are ropes that hold docked ships in place, and a riding-light is a light that a ship shines when it is anchored.

When little lights in little ports come out,
Quivering down through water with the stars,
And all the fishing fleet of slender spars
Range at their moorings, veer with tide about;

When race of wind is stilled and sails are furled,
And underneath our single riding-light
The curve of black-ribbed deck gleams palely white,
And slumbrous waters pool a slumbrous world;

—Then, and then only, have I thought how sweet
Old age might sink upon a windy youth,
Quiet beneath the riding-light of truth,
Weathered through storms, and gracious in retreat.

Which choice best states the main idea of the text?

- A. The tranquility of a port in the evening can incline a person to appreciate the stillness of old age.
- B. The difficulty of bringing a ship into port is apt training for dealing with the types of struggles encountered in old age.
- C. A person who leads a long life that is varied and active may find it difficult to stay in a calm place.
- D. The contrast between the peacefulness of a port at night and its activity during the day reflects the contrast between the calm of old age and the vibrancy of youth.

Question 18

The fynbos shrubland is a diverse habitat found only in South Africa. It is adjacent to the Afro-temperate forest, with almost no transition space between the two distinct habitats. Plants in the fynbos have uniquely thin and long root systems that spread out over large distances to absorb nutrients from the soil. Ecologists transplanted tree seedlings from the forest into plots in the fynbos. Seedlings in plots isolated from the roots of fynbos plants exhibited a growth rate five times greater than that of the seedlings in plots in close proximity to the roots of fynbos plants.

Based on the text, what role do fynbos roots most likely have in maintaining the border between the fynbos shrubland and the Afro-temperate forest habitats?

- A. Fynbos roots damage the root systems of forest plants, leaving those plants unable to acquire sufficient nutrients.
- B. Fynbos roots extend close enough to the forest plants' roots that they constitute a physical barrier that forest plants' roots cannot pass.
- C. The root systems of fynbos plants allow the plants to take in so many soil nutrients that forest plants are prevented from flourishing in the fynbos.
- D. The root systems of fynbos plants enhance the soil immediately surrounding the plants, allowing them to thrive in an otherwise harsh habitat.

Question 19

Jürgen Kocka and other historians of capitalism rarely discuss domestic capitalism in Africa before the period of European colonization, implicitly presenting capitalism as external to and imposed on Africa. Crislayne Alfagali and other Africanist scholars have shown, however, that in parts of Africa, returns-focused investment, the establishment of open markets for wage labor, and other features of capitalism predated colonization. One reason for this discrepancy is that historians of capitalism tend to focus on longitudinal economic data drawn from archival records, which do not exist for much of precolonial Africa.

Which statement about Alfagali and other Africanist scholars is best supported by information in the text?

- A. They likely make use of different types of evidence than historians of capitalism typically rely on.
- B. They likely differ from historians of capitalism in the methods they use to derive longitudinal economic data from archival records.
- C. They likely have a different view about which activities should be considered capitalist in nature than historians of capitalism do.
- D. They likely view capitalism as having been more beneficial for Africa than historians of capitalism do.

Question 20

Philosophers note that many people have an intuitive sense that while we ought not to lie, there may be circumstances in which lying is permissible. If this intuition is correct and we lack an inviolable duty to speak truthfully, what grounds opposition to lying in the first place? Japa Pallikkathayil has advanced one answer by appealing to a duty to respect others' agential interests: the possession of false beliefs constrains agency, and thus we ought not to impede the formation of true beliefs unless doing so prevents a greater constraint on someone's agency or an otherwise impermissible end.

Which choice best states the main idea of the text?

- A. One potential means of justifying opposition to lying is Pallikkathayil's argument that we have an obligation to respect other people's agency that entails a commitment to truthfulness except in certain circumstances.
- B. Many people have an intuitive sense that lying is permissible in some circumstances but lack a principled way to identify those circumstances, and Pallikkathayil's argument may provide a means of resolving that problem.
- C. Pallikkathayil's argument suggests that if we have a duty to respect other people's agential interests and if possession of false beliefs constrains agency, then we have an inviolable duty to speak truthfully.
- D. Pallikkathayil's argument shows that if our intuition that circumstances may make lying permissible is correct, then it is unclear whether there are any grounds for an opposition to lying in the first place.

Question 21

In their 2022 paper, Christos Dimopoulos et al., having granted that the existence of antigravity—in which antimatter and matter repel rather than attract each other—lacked affirmative experimental support, rightly argued that such antigravity was worth considering on theoretical grounds given that evidence against it was similarly lacking. But a 2023 report by an international team of researchers details the first direct ballistic observations of antihydrogen atoms under gravity inside a CERN particle accelerator. Corresponding most closely to predictions under gravitational attraction, these observations were thoroughly inconsistent with antigravity.

Which choice best states the main idea of the text?

- A. Antihydrogen ballistics observations were conducted at CERN to test specific conclusions about antigravity presented in the 2022 paper by Dimopoulos et al.
- B. Although theoreticians were justified in studying antigravity before the release of the 2023 report, the report's findings suggest that the rationale for theoretical consideration offered in the 2022 paper by Dimopoulos et al. is no longer applicable.
- C. The theoretical approach represented in the 2022 paper by Dimopoulos et al. assumed that unambiguous proof of antigravity would not be achievable, but the results in the 2023 report undermine that assumption.
- D. Before 2023, researchers' inordinate focus on theoretical considerations hindered the development of the experimental regimen for direct antihydrogen ballistics observations.